UADO: Uncertainty-Aware Dispatch Optimization for Roadside Assistance via Expected-Cost Minimization and Bayesian Feedback

***Abstract*—Roadside-assistance dispatch becomes a decision problem under uncertainty when driver-reported symptoms do not uniquely identify the required service and providers differ in capability. This paper presents UADO, a three-stage framework that infers a probability distribution over 19 vehicle-fault classes using an adaptive questionnaire with optional OBD-II telemetry, selects providers through capability-aware expected-cost optimization over the full diagnostic distribution, and adapts recurring symptom-pattern beliefs using Bayesian outcome feedback. Evaluation on real holdout data achieves 64.4% questionnaire-only and 69.0% questionnaire-plus-OBD top-1 accuracy, while calibration analysis identifies overconfidence in the decision-tree probabilities. In a 30-seed matched-diagnosis ablation, full-distribution dispatch increases capable-provider assignment from 95.0% to 97.0% and reduces re-dispatch from 5.1% to 3.0%, a 41% relative reduction, compared with argmax dispatch. Under controlled distribution shift, Bayesian feedback further improves exact-type match by 45.4% and reduces re-dispatch by 47.4% relative to expected-cost optimization alone. Sensitivity analysis preserves both conclusions across all tested configurations while identifying the tree-to-history weight as the most influential adaptation parameter. These results demonstrate the operational value of preserving diagnostic uncertainty and incorporating outcome feedback in capability-aware roadside dispatch.**

***Index Terms*—roadside assistance, uncertainty-aware dispatch, vehicle fault diagnosis, expected-cost optimization, Bayesian adaptation**

I. Introduction

Vehicle breakdowns are stressful, time-critical, and geographically unpredictable events. Commercial roadside-assistance services typically dispatch by proximity, provider rating, or a simple weighted score.

We identify three sources of uncertainty that naive dispatch systems ignore. **Diagnostic uncertainty**: a driver's report rarely identifies the fault with certainty, so a dispatch system should reason over a distribution across candidate faults rather than commit early to a single most-likely guess. **Capability mismatch**: providers are heterogeneous (a locksmith cannot tow, a mobile mechanic cannot perform heavy recovery), and dispatching an incapable provider costs a re-dispatch penalty of tens of minutes. **Distributional drift**: any diagnostic model is trained on a finite historical dataset, and the true field population of faults evolves; a system that never recalibrates from outcomes will silently degrade.

UADO (Uncertainty-Aware Dispatch Optimization) is a three-stage pipeline addressing all three: a *Diagnostic Triage* stage producing a full probability distribution over fault classes from an adaptive questionnaire and optional OBD-II telemetry; an *Expected Cost Minimization* (ECM) optimizer that selects a provider by integrating dispatch cost over the entire distribution rather than its argmax; and a *Bayesian Feedback Loop* that updates per-symptom-pattern posteriors from post-resolution outcomes. Frequent, low-severity emergencies (lockouts, flat tyres, empty tanks) short-circuit the learned pipeline via a deterministic fast path, reserving the diagnostic model for genuinely ambiguous cases.

This study is organized around three research questions, each isolated by a dedicated, controlled ablation:

- **RQ1:** Does questionnaire-plus-OBD triage improve diagnostic performance over questionnaire-only triage on real holdout data, and how well calibrated are the resulting probabilities?
- **RQ2:** Holding diagnosis and cost formulation fixed, does dispatching on the full diagnostic distribution improve operational allocation over collapsing that distribution to a one-hot argmax?
- **RQ3:** Under a controlled training-to-field distribution shift, does Bayesian outcome feedback improve dispatch relative to expected-cost optimization without feedback?

The contribution is not a new classifier, conjugate Bayesian model, or general optimization theorem. It is an application-specific integration and controlled evaluation of: (1) interpretable probabilistic triage over 19 learned fault classes with explicit calibration metrics; (2) a capability-aware expected-cost dispatch objective whose operational benefit over argmax dispatch is isolated through a matched-diagnosis ablation; (3) a discounted Dirichlet-multinomial feedback layer validated end-to-end in the application runtime; and (4) a 30-seed, paired discrete-event simulation with two independent ablation arms, plus a sensitivity sweep, that separately attribute gains to full-distribution reasoning and to outcome feedback.

II. Related Work

A. Vehicle Fault Diagnosis

Automated vehicle fault diagnosis has progressed through hand-authored expert systems [2], [3], case-based reasoning [1], [4], fuzzy fault-tree analysis [5], rough-set theory [7], and neural ensembles that generalize well at the cost of interpretability [6]. Decision trees with structured knowledge bases remain favoured where interpretability matters [8], and OBD-II's standardization has enabled parallel telemetry-augmented work [9]. Across this literature, diagnosis is the terminal

output; none of these systems couple it to a downstream dispatch decision.

B. Dispatch Optimization Under Uncertainty

The closest published analogue is the emergency medical services (EMS) literature, where ambulance selection is formulated as stochastic or robust optimization over uncertain demand and travel time [11], [12], but the requested service type is consistently assumed known. More directly, Buil *et al.* formulate real-time roadside-incident allocation using resource skills, priorities, cost, and response time [10], showing roadside allocation is itself a meaningful optimization problem; in their formulation, however, the service requirement is given rather than inferred under uncertainty.

C. Roadside Assistance Systems and Gap

Commercial and academic roadside-assistance platforms focus on the communication and logistics layer (location sharing, SOS alerting, proximity- or rating-based suggestion) without modelling diagnostic uncertainty or outcome feedback [13]. The gap UADO addresses is therefore narrower than "uncertainty-aware dispatch" generally: the service requirement itself is uncertain. No system in our review couples diagnostic-uncertainty-aware triage, a capability-mismatch-penalizing cost function, and a self-correcting posterior update; we make no claim that any individual constituent method is independently novel.

III. System Architecture

Fig. 1 shows the UADO pipeline. A driver-reported incident enters at Diagnostic Triage. If the intent matches one of ten deterministic fast-path categories (lockout, flat tyre, empty tank), the pipeline short-circuits directly to dispatch: no tree inference, no Bayesian correction. Otherwise, an adaptive questionnaire (and OBD-II telemetry if available) passes through the decision tree, producing a probability distribution over 19 fault classes, which is blended with any learned posterior before reaching the ECM optimizer. The optimizer ranks providers by expected cost and selects the minimizer. After resolution, the confirmed outcome feeds the Bayesian Feedback Loop, updating the posterior and provider trust score for future dispatches.

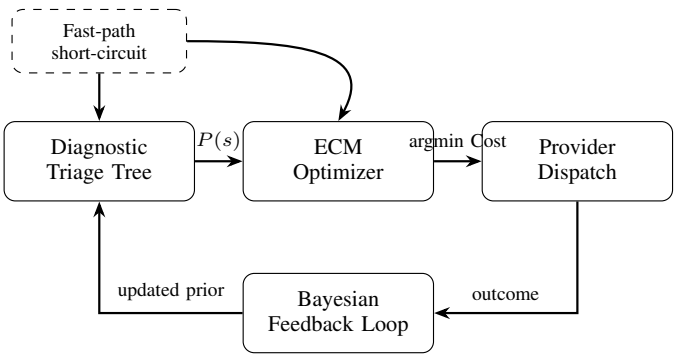


Fig. 1. UADO three-stage pipeline. Fast-path intents bypass tree inference entirely; the ML path blends tree output with a learned prior before dispatch; post-resolution feedback closes the loop.

The runtime distinguishes 29 total service types: 19 ML-diagnosable fault classes and 10 fast-path types. Five provider types (mobile mechanic, fuel delivery, locksmith, light tow, heavy tow) are matched via a capability matrix; a provider's mismatch risk is the total probability mass on service types it cannot perform, the quantity ECM penalizes.

IV. Methodology

A. Diagnostic Triage

The triage stage operates in two tiers. *Tier 1* (questionnaire-only) uses categorical answers from an adaptive branching form: an entry-point intent question routes to follow-up chains (e.g., cranks-but-no-start branches into sound produced, each diagnostically distinct). Skipped questions are encoded as explicit `NOT_ASKED` rather than missing, since the tree treats non-applicability as diagnostic information. *Tier 2* (OBD-enhanced) additionally incorporates continuous OBD-II telemetry (battery voltage, coolant temperature, RPM) when available.

Both tiers use a `scikit-learn` decision tree (Gini impurity, `random_state=42`, `min_samples_leaf=1`) exported as nested JSON and walked at request time. Each leaf stores a full class-probability distribution $P_{\text{tree}}(s)$, and confidence is reported as normalized entropy, $1 - H(P)/\log_2(19)$. Because expected-cost dispatch consumes probabilities rather than only labels, we evaluate both discrimination (top-1, top-3, macro-F1) and probability quality (multiclass Brier score, log loss, expected calibration error). A 200-tree Random Forest is retained as a model-selection comparison but is not used in the application runtime dispatch path.

B. Expected Cost Minimization (ECM) Optimizer

The simulator and application runtime use the same provider-cost formulation. For each candidate provider p, given the diagnostic probability distribution P over service types s, the expected operational cost is

$$\text{Cost}(p) = \frac{t_{\text{travel}}(p) + \sum_s P(s)\left[t_{\text{svc}}(s) + M(p,s)\right] + \lambda\left(\frac{\tau(p)}{10}\right) t_{\text{travel}}(p)}{\text{trust}(p)} \quad (1)$$

where $t_{\text{travel}}(p)$ is the travel time to provider p, $t_{\text{svc}}(s)$ is the average service duration for service type s, $\tau(p) \in [1, 10]$ is the traffic-impact score, $\lambda = 0.3$, and $\text{trust}(p) \in [0.1, 1.0]$ is the provider trust score. The mismatch cost is defined as

$$M(p,s) = \begin{cases} t_{\text{assess}} + R, & p \text{ cannot perform } s, \\ 0, & \text{otherwise}, \end{cases} \quad (2)$$

where $t_{\text{assess}} = 10$ min represents the assessment delay and $R = 45$ min is the baseline re-dispatch penalty, giving a total mismatch cost of 55 min. The selected provider is

$$p^* = \arg\min_p \text{Cost}(p). \quad (3)$$

The traffic weight λ and mismatch-cost components are design parameters. Section VI-D evaluates sensitivity to the

re-dispatch penalty R, discount factor γ, and tree-to-history weight K.

The distribution term in Eq. (1) penalizes a provider proportionally to the probability mass on faults it cannot perform; a provider capable of covering the two or three most probable fault types can resolve an incident correctly without matching the argmax class exactly. To isolate this effect (RQ2), we define a controlled one-hot ablation that keeps the same incidents, diagnostic predictions, providers, and cost function, but replaces $P(s)$ with

$$P_{\text{1hot}}(s) = \mathbf{1}\left[s = \arg\max_j P(j)\right]. \tag{4}$$

Any difference in dispatch outcome between the two arms is therefore attributable to whether uncertainty is retained by the decision layer, not to a different diagnosis.

C. Bayesian Feedback Loop

Each incoming triage response is reduced to a canonical symptom key, a hash of the five highest-Gini-importance features, so recurring patterns accumulate feedback across incidents. For each key x, we maintain a Dirichlet pseudo-count vector $\boldsymbol{\alpha}_x = (\alpha_{x,1}, \ldots, \alpha_{x,19})$, initialized to a weak uniform prior $\alpha^{(0)}_{x,k} = \alpha_0$. On each feedback event reporting fault type k^*, the count vector updates under exponential discounting:

$$\alpha_{x,k} \leftarrow \gamma\, \alpha_{x,k} + \mathbb{1}[k = k^*], \quad \forall k \tag{5}$$

with discount factor $\gamma \in (0, 1]$, giving posterior mean $P_{\text{hist}}(k \mid x) = \alpha_{x,k} / \sum_j \alpha_{x,j}$. At $\gamma = 1$, this corresponds to the standard Dirichlet-multinomial update; for $\gamma < 1$, discounting gives greater weight to recent observations and enables adaptation under distribution shift [14]. We adopt $\gamma = 0.99$ and $\alpha_0 = 0.5$ per class, with the effect of discounting evaluated empirically in Section VI-C. At inference, after a minimum of $n_x = 3$ real feedback observations, the tree's prediction blends with the historical distribution via a fixed effective tree weight $K = 20$:

$$P_{\text{final}} = \frac{K \cdot P_{\text{tree}} + n_x \cdot P_{\text{hist}}}{K + n_x}. \tag{6}$$

Below the $n_x = 3$ threshold, blending is suppressed entirely to prevent a single noisy report from steering dispatch. Equation (6) is an engineering weighting rule rather than a property implied by the discounted Dirichlet model; Section VI-D reports its sensitivity to K directly, since K turns out to be the most influential of the tested parameters. Diagnostic uncertainty is reported alongside every prediction as Shannon entropy, $H(P) = -\sum_k P(k) \log_2 P(k)$.

V. Experimental Setup

A. Dataset, Splitting, and Augmentation

Training data comprises 443 field-collected questionnaire responses, logic-checked against a structured automotive fault knowledge base and reviewed by expert mechanics, and 285 real OBD-II sensor readings (15 per class across 19 classes) labelled with resolved service type. Nine rows (2.0%) belonging to two fault classes absent from the current runtime taxonomy were dropped. High-cardinality contextual attributes unrelated to fault diagnosis were excluded before training.

Real holdout observations are separated *before* augmentation. To address class imbalance and limited training volume, class-conditional augmentation was applied only to the training partition. Questionnaire records were augmented using template-based generation, where each synthetic row starts from a verified real row and only vehicle/context fields are resampled while diagnostic fields are retained. OBD readings were generated by class-conditional resampling from $\mathcal{N}(\mu_{\text{class}}, 0.9\sigma_{\text{class}})$. Synthetic rows were capped at $3\times$ the real count per class, yielding 989 questionnaire and 950 OBD training records. No synthetic record was used for evaluation.

For Tier 2, each questionnaire row is paired with a randomly or distribution-sampled OBD row sharing its resolved service type, since the two sources are not jointly observed incidents; this class-matched pairing is disclosed as a limitation in Section VII and means Tier 2 results should be read as proof-of-concept evidence that telemetry adds signal, not as a deployment-accuracy estimate.

A planned enhancement, OBD-II diagnostic trouble codes (DTCs) as a Tier 2 feature grounded in a 183-entry expert-curated reference, was excluded from training after an audit found severe label leakage: 108 of 120 candidate DTCs mapped to exactly one class by construction, which would have inflated holdout accuracy by an estimated 15–30 pp from leakage alone. Excluding these codes is treated as a successful validation control rather than a setback: it prevented an inflated diagnostic-feature claim from reaching the reported results.

Alongside the application runtime `DecisionTreeClassifier`, we train a `RandomForestClassifier` (200 trees, identical hyperparameters otherwise) as a model-selection comparison arm.

B. Discrete-Event Dispatch Simulation

We built a discrete-event simulation of 1,000 incidents against 20 heterogeneous providers distributed uniformly over a metropolitan bounding box, with Poisson-process arrivals (rate 0.5/min). Compared strategies within a seed observe the identical incident stream in the identical order, enabling per-incident paired statistical tests rather than weaker independent-sample comparisons. Thirty seeds are evaluated; we report mean $\pm$ SD and, for match rate, McNemar's test with continuity correction, and for resolution time, a paired t-test.

For the RQ3 feedback ablation, ground truth is deliberately drawn from a perturbed class-frequency distribution ($2\times$ the trained rate for one fault class, $0.5\times$ for another, $1.5\times$ for a third, renormalized over 19 classes) rather than the tree's training distribution, so the tree alone reflects only the stale distribution and the Bayesian layer has a genuine gap to close from outcomes alone. Questionnaire answers are sampled conditional on this ground truth with 15% per-field response

noise. RQ2 (full-distribution vs. one-hot ECM, Eq. (4)) is run as a separate, matched-diagnosis comparison so the two mechanisms are never conflated in one result.

Compared strategies are:

- **GreedyNearest**: nearest available provider regardless of capability.
- **TypeMatched**: tree's argmax prediction only, nearest capable provider for that type.
- **MultiCriteria**: weighted score $0.4\times\frac{1}{1+t_{\text{travel}}}+0.3\times\text{trust}+0.3\times\mathbb{1}[\text{capable}]$ against the argmax prediction, no learning.
- **ECM-only**: full-distribution ECM (Eq. (1)) with Bayesian feedback disabled — the RQ3 comparator.
- **One-hot ECM**: full-distribution ECM with $P(s)$ replaced by Eq. (4) — the RQ2 comparator.

Resolution time is modelled as $t_{\text{travel}} + t_{\text{svc}}(\text{true type}) + \mathbb{1}[\text{incapable}] \cdot 55\,\text{min}$, consistent with Eq. (1).

C. Sensitivity Design

To check that the RQ2/RQ3 conclusions are not artifacts of the specific parameter values in Eqs. (1)–(6), we perform a one-factor-at-a-time sensitivity analysis over $R \in \{30, 45, 60\}$ min, $\gamma \in \{0.95, 0.99, 1.00\}$, and $K \in \{10, 20, 40\}$, with $(R = 45, \gamma = 0.99, K = 20)$ as the baseline. This is a robustness check rather than a tuning search.

VI. Results and Discussion

A. RQ1: Diagnostic Accuracy and Calibration

TABLE I
REAL-HOLDOUT DIAGNOSTIC PERFORMANCE (DT = DEPLOYED-PATH DECISION TREE; RF = RANDOM FOREST COMPARISON ARM)

Tier	Model	Top-1	Top-3	F1	Brier
Q-only	DT	64.4%	65.5%	0.70	0.718
Q-only	RF‡	73.6%	96.6%	0.78	—
Q+OBD	DT	**69.0%**	69.0%	0.74	0.621
Q+OBD	RF‡	**81.6%**	**97.7%**	**0.86**	—

‡Demonstrated accuracy ceiling; not yet integrated into the application runtime dispatch path (Section VII). DT log loss / ECE: 12.85 / 0.362 (Q-only), 11.19 / 0.310 (Q+OBD).

Random guessing over 19 classes yields 5.3% top-1 accuracy and always predicting the majority class yields 15.0%; all configurations substantially exceed both trivial baselines. Adding class-matched OBD telemetry improves the application runtime tree from 64.4% to 69.0% top-1 accuracy and from F1 0.70 to 0.74. Random Forest outperforms the application runtime tree by a wide and consistent margin (+9.2 pp Tier 1, +12.6 pp Tier 2). We report this as a model-selection comparison rather than replacing the evaluated decision-tree pipeline, since the tree provides single-path interpretability and Random Forest integration would require separate runtime validation.

The calibration metrics are the more consequential result for dispatch, since ECM operationally weights $P(s)$ rather than only the argmax label. Both tiers show high ECE (0.362 and 0.310), consistent with an unregularized tree whose small or pure leaves assign extreme probabilities to observations that do not generalize. Poor absolute calibration does not invalidate the distribution as a ranking signal. The RQ2 ablation below tests this directly, but it does mean raw leaf probabilities should not yet be read as well-calibrated confidence estimates, and explicit calibration (e.g., Platt scaling or isotonic regression) is identified as near-term future work rather than treated as already solved.

As a concrete illustration, four classes sharing elevated coolant temperature are disambiguated only by driver context: an OBD-only model predicted entirely outside this fault family in 7/19 holdout cases, versus 1/19 for the combined tier, showing that a shared sensor effect needs situational context the questionnaire, not telemetry, supplies.

B. RQ2: Does Preserving the Full Distribution Improve Dispatch?

TABLE II
CONTROLLED ECM REPRESENTATION ABLATION, 30 SEEDS (MATCHED DIAGNOSIS, UNIFIED COST FORMULATION)

ECM input	Match	Capable	Re-disp.	Res. (min)
Full distribution	34.4%	**97.0%**	**3.0%**	**67.9**
Argmax / one-hot	34.4%	95.0%	5.1%	68.6

Table II holds diagnosis and cost formulation fixed and varies only how the distribution is consumed. Exact-type match is identical (34.4%) in both arms, as expected, since the underlying diagnostic argmax is unchanged. Operational outcomes differ: retaining the full distribution raises capable-provider assignment from 95.0% to 97.0% and cuts re-dispatch from 5.1% to 3.0%, a relative reduction of approximately 41% (secondary pooled McNemar test on the capable outcome, $\chi^2 = 584.7$, $p < 0.0001$). This is the direct evidence for the paper's central decision-layer claim: the distribution term in Eq. (1) penalizes a provider when non-argmax probability mass falls on services it cannot perform, and one-hot dispatch discards that information entirely. Preserving diagnostic uncertainty improves capability matching even though the classifier's probabilities are, per Section VI-A, imperfectly calibrated — the ranking information in the distribution is useful independent of its absolute calibration quality.

C. RQ3: Marginal Contribution of Bayesian Feedback

TABLE III
FEEDBACK ABLATION UNDER CONTROLLED DISTRIBUTION SHIFT, 30 SEEDS × 1,000 INCIDENTS

Strategy	Match	Capable	Re-disp.
UADO	**34.4%**	**97.0%**	**3.0%**
ECM-only	23.7%	94.3%	5.7%
GreedyNearest	25.1%	71.0%	29.0%
TypeMatched	23.7%	94.2%	5.8%
MultiCriteria	23.7%	94.0%	6.0%

Table III isolates the Bayesian layer's contribution: UADO and ECM-only use identical cost optimization, differing only in whether Dirichlet blending is enabled. Under the unified cost formulation, feedback alone contributes an approximately 45.4% relative improvement in match rate (34.4% vs. 23.7%) and an approximately 47.4% relative reduction in re-dispatch (3.0% vs. 5.7%) on top of expected-cost optimization without feedback. Against GreedyNearest, the reduction in re-dispatch is far larger still, since nearest-provider selection ignores capability entirely. The match-rate difference remains significant under a paired McNemar test ($p < 0.0001$). Resolution-time differences were evaluated separately using paired tests.

ECM-only, TypeMatched, and MultiCriteria converge to an identical 23.7% match rate in this table. RQ2 (Table II) resolves what would otherwise be an open question about this convergence: under matched diagnosis, full-distribution ECM and one-hot dispatch produce the same argmax-driven match rate but diverge sharply in capable-provider and re-dispatch outcomes, indicating that the capability-mismatch penalty empirically favours the argmax-consistent provider type for match purposes while still using the full distribution to avoid incapable providers. Only UADO, which corrects tree errors from accumulated feedback, escapes the shared match-rate ceiling that ECM-only, TypeMatched, and MultiCriteria share.

D. Bayesian Convergence Diagnostics

The discounted Dirichlet-multinomial update (Eq. (5)) is validated end-to-end in the application runtime (28/28 unit tests). Under 200 clean observations ($\gamma = 1$), the posterior concentrates 95.7% of its mass on the true class (uniform prior $\approx 5.3\%$). Under 30% injected label noise, the MAP class remains correct throughout. Under a simulated shift (true class switches at observation 300), the discounted model ($\gamma = 0.99$) favours the new class 78.6% of the time by 150 observations post-shift, while a non-discounting control ($\gamma = 1$) remains dominated by the stale class (65.4%), evidence that discounting, not posterior consistency alone, is the right design choice for tracking real-world drift.

E. Sensitivity Analysis

TABLE IV
SENSITIVITY OF UADO TO KEY DISPATCH AND FEEDBACK PARAMETERS

Configuration	Match	Capable
Baseline (R=45, γ=.99, K=20)	34.4%	97.0%
$R = 30$	34.4%	96.5%
$R = 60$	34.4%	97.3%
$\gamma = 0.95$	35.6%	97.1%
$\gamma = 1.00$	34.1%	97.0%
$K = 10$	**41.1%**	**97.4%**
$K = 40$	27.8%	96.5%

Table IV varies one parameter at a time around the baseline. The strategy orderings established in RQ2 and RQ3 hold in all seven configurations: UADO exceeds ECM-only in match rate, and full-distribution ECM exceeds one-hot ECM in capable-provider assignment, in every case. Varying R and γ changes outcomes only modestly. The tree-to-history weight K is substantially more influential: $K = 10$ raises match rate to 41.1%, while $K = 40$ lowers it to 27.8%, since a lower K gives accumulated feedback proportionally more influence over Eq. (6), adapting faster under the imposed shift. We do not read this as evidence that $K = 10$ is globally optimal; only three values were tested under an experiment built to reward fast adaptation, but it identifies K as the parameter meriting data-driven tuning while confirming the comparative conclusions hold regardless of the baseline chosen.

F. Implications

RQ2 shows that consuming the full diagnostic distribution rather than its argmax measurably improves capability-aware provider selection under a fixed, imperfectly calibrated classifier. RQ3 shows that closing the loop with post-resolution outcomes recovers accuracy lost to drift, purely from feedback, without changing the underlying model. UADO's 34.4% exact-type match versus 97.0% capable-provider rate reflects that a provider covering the two or three most probable fault types resolves an incident correctly without matching the single most-likely class.

VII. LIMITATIONS AND FUTURE WORK

The present evaluation is a simulation-backed proof of concept rather than a field deployment study. Diagnostic evaluation uses real holdout observations separated before augmentation; however, questionnaire and OBD-II data were collected independently and are class-matched for Tier 2 evaluation, so the multimodal result should be interpreted as preliminary evidence until jointly observed incidents are available. Dispatch experiments use controlled provider, traffic, and distribution-shift assumptions and therefore establish comparative rather than deployment-level performance. Sensitivity analysis preserves the principal strategy ordering across all tested configurations, although the tree-to-history weight K materially affects adaptation performance. Future work will investigate data-driven or adaptive selection of K, explicit probability calibration, jointly collected questionnaire–OBD observations, and prospective evaluation using real roadside incidents and provider outcomes.

VIII. CONCLUSION

This paper presented UADO, a three-stage dispatch pipeline treating diagnostic uncertainty, provider capability mismatch, and distributional drift as first-class problems. A decision-tree triage stage produces a full probability distribution over 19 fault classes, reaching 64.4 - 69.0% real-holdout accuracy, a Random Forest comparison demonstrates a further 9–13 pp ceiling not yet in application runtime. Using a consistent ECM formulation across the simulator and application runtime, a matched-diagnosis ablation shows that full-distribution dispatch reduces re-dispatch by approximately 41% relative

to one-hot dispatch ($5.1\% \to 3.0\%$) while raising capable-provider assignment from 95.0% to 97.0% ($\chi^2 = 584.7$, $p < 0.0001$). An independent, controlled-drift ablation shows Bayesian feedback, validated end-to-end in application runtime (28/28 unit tests), raises match rate by approximately 45.4% and cuts re-dispatch by approximately 47.4% relative on top of expected-cost optimization alone ($p < 0.0001$). A sensitivity sweep over the mismatch penalty, discount factor, and tree-to-history weight shows both ablations' conclusions hold across all seven configurations, identifying the tree-to-history weight K as the parameter most needing future tuning. A label-leakage audit (108/120 candidate DTC codes unique to one class) excluded a planned feature rather than report an inflated result. Priorities for future work are data-driven selection of K, explicit calibration, jointly collected questionnaire–OBD data, and a field trial with real provider outcomes.

REFERENCES

[1] S. De Sumana and B. Chakraborty, "Case based reasoning (CBR) methodology for car fault diagnosis system (CFDS) using decision tree and Jaccard similarity method," in *Proc. 2018 3rd Int. Conf. for Convergence in Technology (I2CT)*, 2018, pp. 1–6.

[2] P. Chen and Y. Min, "Research and implementation of vehicle fault diagnosis expert system," *Int. J. Multimedia and Ubiquitous Engineering*, vol. 5, no. 7, 2013.

[3] S. A. Mostafa, M. S. Ahmad, M. A. Mohammed, and O. I. Obaid, "Implementing an expert diagnostic assistance system for car failure and malfunction," *Int. J. Computer Science Issues*, vol. 9, no. 2, pp. 1–8, Mar. 2012.

[4] Z. Wen, J. Crossman, J. Cardillo, and Y. L. Murphey, "CBR in vehicle fault diagnosis," in *Proc. Int. Joint Conf. on Neural Networks*, vol. 4, 2003, pp. 2679–2684.

[5] M. Sharma, R. Bhardwaj, and R. Shrivastava, "Car failure fuzzy fault diagnostic system based on fault tree analysis," in *Proc. Int. Conf. on Industrial and Information Systems*, 2010, pp. 18–21.

[6] Y. L. Murphey, Z. Chen, M. Abou-Nasr, R. Baker, T. Feldkamp, and I. Kolmanovsky, "Ensembles of neural networks with generalization capabilities for vehicle fault diagnosis," in *Proc. Int. Joint Conf. on Neural Networks*, 2009, pp. 2188–2194.

[7] H. Zhang, D. Jin, and X. J. Zhao, "Research on extracting vehicle fault symptoms based on rough set theory in mechanical engineering," *Applied Mechanics and Materials*, pp. 110–114, 2013.

[8] A. Pérez-Vázquez, M. Anzures-García, and L. A. Sánchez-Gálvez, "Vehicle engine fault diagnosis approach based on a decision tree and knowledge base," *Int. J. Combinatorial Optimization Problems and Informatics*, vol. 15, no. 2, pp. 185–197, 2024.

[9] E. T. Michailidis, A. Panagiotopoulou, and A. Papadakis, "A review of OBD-II-based machine learning applications for sustainable, efficient, secure, and safe vehicle driving," *Sensors*, vol. 25, no. 13, p. 4057, 2025.

[10] R. Buil, J. de Armas, D. Riera, and S. Orozco, "Optimization of the real-time response to roadside incidents through heuristic and linear programming," *Mathematics*, vol. 9, no. 16, Art. no. 1982, 2021.

[11] V. Guigues, "Operation of an ambulance fleet under uncertainty," Optimization Online, 2022.

[12] D. Bertsimas and Y. Ng, "Robust and stochastic formulations for ambulance deployment and dispatch," *European Journal of Operational Research*, vol. 279, no. 2, pp. 557–571, 2019.

[13] D. D. Nigade, A. N. Atole, V. S. Potare, and R. D. Bobade, "On-road vehicle breakdown assistance finder," *International Research Journal of Modernization in Engineering Technology and Science*, vol. 7, no. 11, pp. 7716–7724, 2025.

[14] M. West and J. Harrison, *Bayesian Forecasting and Dynamic Models*, 2nd ed. New York, NY, USA: Springer-Verlag, 1997.

[15] D. Bertsimas and J. N. Tsitsiklis, *Introduction to Linear Optimization*. Belmont, MA, USA: Athena Scientific, 1997.